

Manual

Connection of KABA Offline Components ZKS-SOK 8.2

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1 Overview: Connection of Offline Components

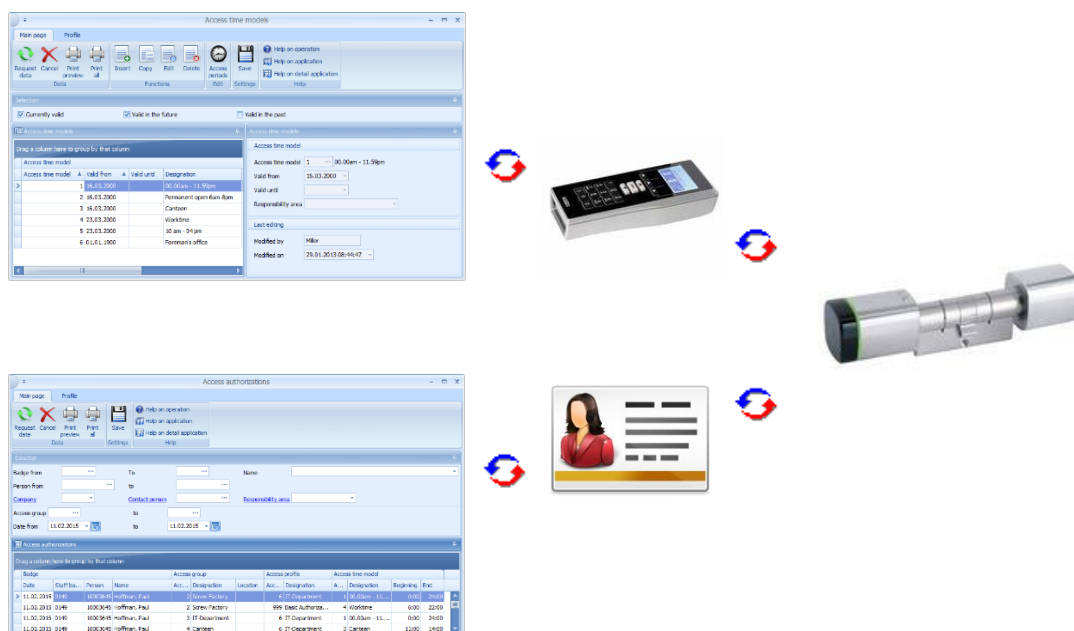
Purpose

This function package includes an interface to connect Kaba GmbH offline components to HYDRA Access Control. Since these components are battery-operated and their authorizations are transported via the badge, no structural changes are required for routing supply or data lines.

The access configurations (access time models, public holidays, ...) and the reading of the access logs are synchronized via the Kaba Programmer 1460, which is synchronized via the Kaba software B-COMM. Changes in configurations are automatically transferred from HYDRA to B-COMM via an interface.

The authorizations for the offline components are written on the employee badge at PZE terminals with the terminal program AIP or ctwin. Since the PZE terminals are permanently connected to HYDRA, changes in the authorizations can be loaded promptly on the employee's badge, without the administrator having to load the new authorizations on all offline components concerned.

The following illustration shows the data flow using the example of a cylinder:



Implementation notes

You use the function package if:

- you wish to control and monitor door access without routing a communication line and/or voltage supply to the relevant door;
- you intend to substitute a locking system by access readers;

- the HYDRA Access Control does not require a current status (open, closed, disturbed, ...) of the relevant doors.

Integration

This function package can only be used if Access Control is used in HYDRA (function package *Management Functions Access Control*). The connection is available for the following offline components:

- Digital cylinder of the *Kaba evolo* type
- Electronic door mounts of the *Kaba c-lever* and *Kaba c-lever compact* type

Features

- Connection of offline components
 - Interface for transferring access configurations to B-COMM
 - Writing of authorizations on the badge

Differences between offline door mounts and digital cylinders

The following differences exist between offline door mounts and digital cylinders:

- An offline door mount behaves like a door handle and can therefore only be used to open the door.
- A digital cylinder behaves like a key and can therefore be used to open and lock a door.

Using offline door mounts has the following advantages:

- Simple operation due to the door handle
- An ordinary lock cylinder may be fitted on the door mount

Using digital cylinders has the following advantages:

- The door does not need to be modified (e.g. important for fire doors)
- A digital cylinder can be used if the offline mount is too wide.

2 Installation of Offline Component Connection

2.1 Installation of an update for HYDRA

HYDRA systems initially installed before the third quarter of 2014 usually require an installation procedure on the HYDRA server, even if current Service Packs have been installed.

This installation procedure is used to provide the necessary prerequisites in the HYDRA database.

This subsequent installation is coordinated by the MPDV project management, if required, and implemented with the aid of a separate installation manual.

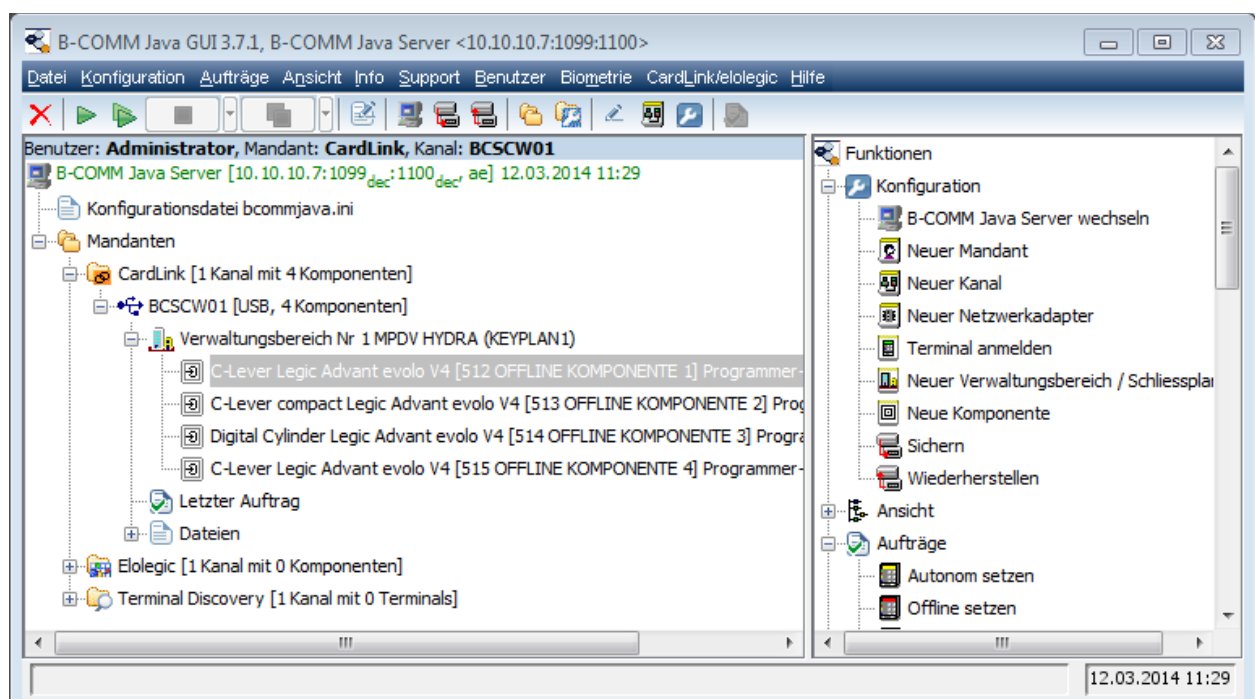


HYDRA systems initially installed from the third quarter of 2014 onwards in the version of Service Pack 5 already meet the requirements on the HYDRA server so that an installation is not necessary there.

2.2 Requirements

- The HYDRA server must have Java version 1.5 as a minimum requirement.
- The KABA B-COMM software (server + GUI) must have been installed. The installation guide for the KABA B-COMM software is included as PDF file on the installation medium.

2.3 Basic configurations in the B-COMM GUI



2.3.1 CardLink instance

A CardLink instance with the name "CardLink" has to be created manually. The option "instance for CardLink" has to be activated. Subsequently, a USB channel named BCSCW01 is automatically created in the "CardLink" instance.

The name "CardLink" for the instance synchronized from HYDRA is fixed.

All settings related to the instance and channel must be managed in B-COMM. HYDRA will not synchronize any data in the instance or channel.

2.3.2 Administration area

An administration area must be created manually in this CardLink instance in B-COMM. By default, HYDRA synchronizes with the administration area with *Number 1*.

The data of the tabs *Parameters* and *Master* must be maintained in the administration area in B-COMM. HYDRA will not synchronize any data in these tabs, since there are no corresponding settings in HYDRA.



When creating an administration area, it must be observed that the *media technology* is set correctly, since it cannot be changed subsequently.

Parameters tab

Number

By default, HYDRA expects an administration area with *Number 1*. Another number can also be used. This must then be taken into consideration when the interface is set up in HYDRA, see below.

Name

Assign a meaningful name.

Validation periods and validation method

The validation period 3 must be configured, e.g. for 1 or 2 days. This validation period will be used by HYDRA for all components and badges.

Other fields

The other fields must be maintained in accordance with the B-COMM documentation.

Master tab

Other fields

The other fields must be maintained in accordance with the B-COMM documentation.



The data in the tabs *Door (groups)* and *Days off/Special days* must not be entered manually in B-COMM, but are maintained in HYDRA and synchronized to B-COMM.

2.3.3 Create component

The components include settings and parameters not known by HYDRA. For this reason, automatic synchronization requires a "copy template" in order to be able to create new components via automatic synchronization. This copy template must be created once in B-COMM upon installation.

Please use door number 512 and the component type you will primarily use for this purpose!

HYDRA will then always use the component with the lowest door number (*Door (groups)* tab, *Door number* field) as copy template, if new accesses not yet existing in B-COMM are created in HYDRA. Please observe that valid door numbers for CardLink start from 512.

The key for synchronizing the HYDRA accesses to the components in B-COMM is the *Door number* managed in the *Door (groups)* tab.



Accesses for Kaba offline components must be created in HYDRA in an access number range from 512 to 4511. Synchronization with B-COMM only takes place for accesses where the *Offline component* option is set.

Details as to which tabs and fields are synchronized upon synchronization from HYDRA can be found in the document dealing with the [configuration of offline components](#).



If you use offline components of different *Types*, it may be necessary to make a manual correction after synchronization, since the type of the component with the lowest number will also be copied.

2.4 Set up the interface to B-COMM in HYDRA

The settings for the interface have to be made in HYDRA. The configurations are managed via the [INI configuration](#).

For this purpose, the INI configuration **CARDLINK** has to be entered first.

Subsequently, a number of [INI data configurations](#) have to be made:

Section BCOMM, key JAVA_INST_PATH

Installation path of Java installation, e.g. "c:\Program Files\Java\jdk1.5.0_16\bin". Setting the parameter is mandatory to ensure correct assignment of the Java version. The path must be indicated without the trailing slash/backslash!

Section BCOMM, key RMI_SERVER

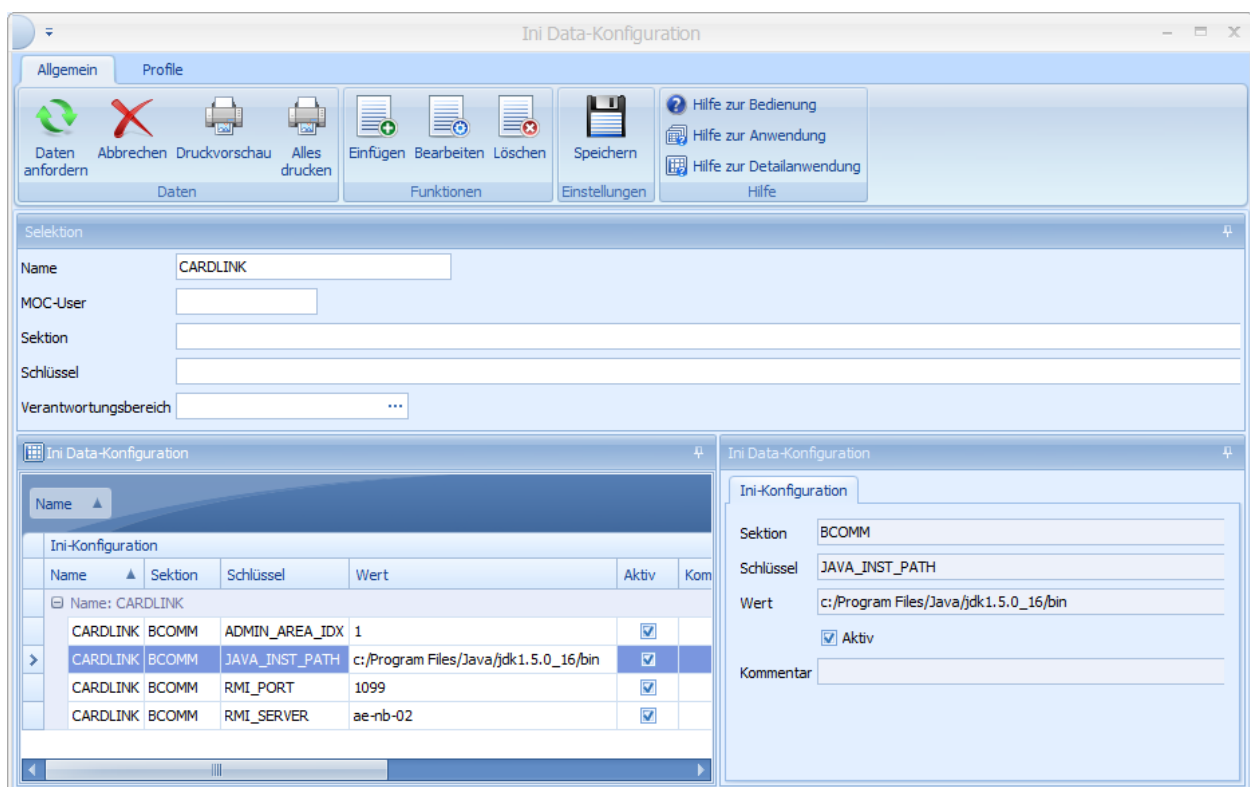
Name or IP address of the computer on which the B-COMM server is installed. Specification is mandatory.

Section BCOMM, key RMI_PORT

RMI port number. Specification is optional, default value is 1099.

Section BCOMM, key ADMIN_AREA_IDX

Number of the administration area managed by HYDRA. Specification is optional, default value is Administration area 1.



2.5 Setup of provision of CardLink data

The authorizations for the access at Kaba offline components are provided cyclically on the HYDRA server and loaded cyclically by the HYDRA PZE terminals in order to write them on the badges of the employees. This configuration only provides authorizations for accesses with KABA offline components. This refers to accesses assigned to a terminal of the type "KABA Programmer".

2.5.1 Configuration of badge type

The type of the badges used for CardLink must be set via an [INI configuration](#). A standardized badge type must be used for each HYDRA system.

For this purpose, the INI configuration **CARDLINK** has to be entered first, if required.

Subsequently, the following [INI data configuration](#) has to be made:

Section OPTIONS, key BADGETYPE

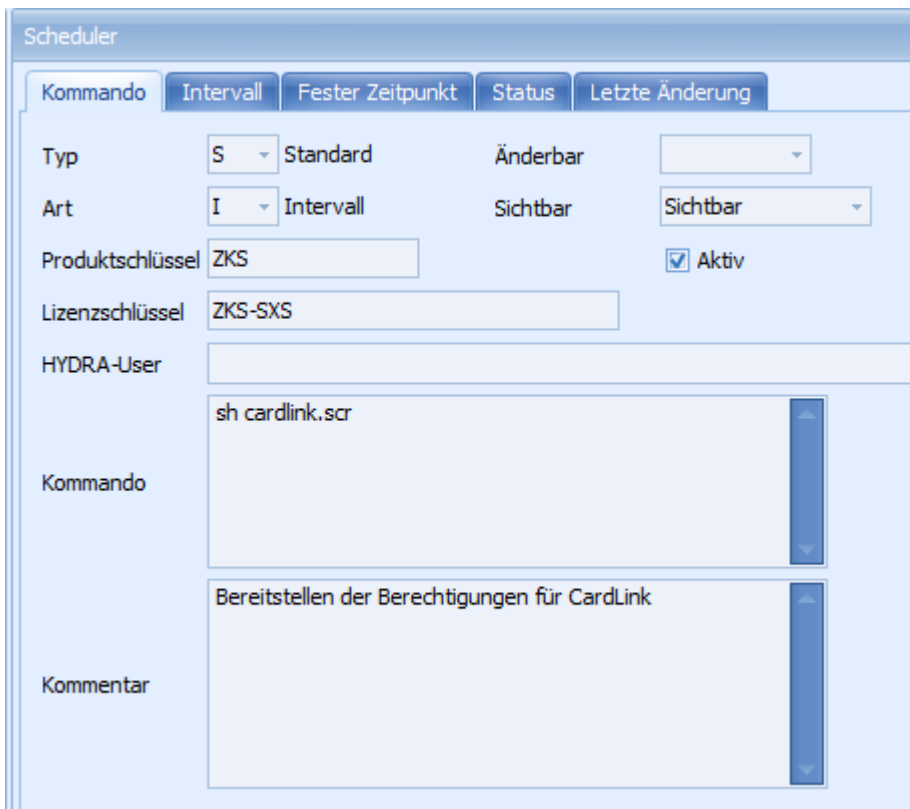
The badge type is indicated as a number:

1 = Legic Prime (default)

2 = Mifare/Legic Advant

2.5.2 Activation in scheduler

The HYDRA server cyclically updates and provides the data required to load authorizations on the badges. For this purpose, an entry must be set up in the Scheduler.



The screenshot shows the 'Scheduler' window with the following configuration:

Kommando	Intervall	Fester Zeitpunkt	Status	Letzte Änderung
Typ	S	Standard	Änderbar	
Art	I	Intervall	Sichtbar	Sichtbar
Produktschlüssel	ZKS		☒ Aktiv	
Lizenzschlüssel	ZKS-SXS			
HYDRA-User				
Kommando	sh cardlink.scr			
Kommentar	Bereitstellen der Berechtigungen für CardLink			

The job is to be set up as 5-minute interval job.

"Comment" tab:

Type	S=Standard
Kind	I=Interval
Visible	Visible
Active	☒ Activated
Product key	ZKS
License key	ZKS-SOK
Command (depending on the server operating system)	
Windows:	sh cardlink.scr
Linux:	cardlink.scr

The administration area can be transferred to the shell script cardlink.scr as a parameter. Multiple administration areas are used, for example, if several productive instances are installed on one HYDRA server.

Comment	Provision of authorizations for CardLink
---------	--

"Interval" tab:

Interval	0:05:00
----------	---------



With the default settings, the authorizations are provided on the server every 5 minutes and loaded to the PZE terminals every 5 minutes, too. So it may take up to 10 minutes with these settings until a modified authorization is available at the PZE-terminal and can be loaded on the badge.

3 Configuration of Kaba Offline Components

3.1 Configurations in HYDRA

3.1.1 Terminals

Since the offline components are not connected to an access terminal in deviation to the online accesses, and since the connection to offline components is made via the programmer, a [terminal](#) of the "Kaba Programmer" type (terminal type 144) has to be created for the offline components. This "terminal" is then entered as such in the accesses created for the offline components.

As regards the PZE terminals to be used for validating and loading authorizations on badges, one of the four absence reason buttons has to be configured with the function "CL" for "Load authorizations".

Writing the authorizations on badges is only possible at terminals with the AIP or ctwin terminal programs.



In order to be able to use a PZE terminal for writing authorizations on badges, it must be fitted with a write-capable reader. In many cases it is necessary to replace the reader in the PZE terminal by a write-capable reader.

3.1.2 Access groups

A maximum of 512 door groups can be created in B-COMM. For this reason, the interface only synchronizes the [access groups](#) with numbers below 512. Access groups with numbers as from 512 are ignored and recorded in the interface log.

Only access groups with a minimum of one KABA offline component are transferred. You can identify these accesses if they are assigned to a terminal of the type "KABA Programmer".

3.1.3 Accesses

The option *offline component* has to be activated for the [accesses](#). Only accesses in the numbering range from 512 through 4511 are considered. The access must be assigned to a terminal of the "Kaba Programmer" type (terminal type 144) so that it is synchronized with B-COMM.



Accesses which are assigned to a terminal of the "Kaba Programmer" type but are not located within the valid range of numbers are ignored and recorded in the error log of the interface.

3.1.4 Access time models / access periods

When configuring the [access time models](#), take into account that B-COMM supports a maximum number of 15 time profiles in one administration area. For this reason, only the access time models with the numbers 1 through 15 are synchronized with B-COMM via the interface.

With B-COMM, times can only be recorded accurately to the minute. The last minute is always included completely in the valid time slot. In HYDRA, a time slot ends with the last second before the indicated end period. For this reason, the end times of the time slots are generally reduced by one minute when transferred to B-COMM. Example: A period from 7:00 AM to 5:00 PM in HYDRA corresponds to a time slot from 7:00 AM to 4:59 PM in B-COMM.

B-COMM only supports times between 12:00:00 AM and 11:59:00 PM in the time slots. HYDRA is theoretically capable of specifying periods with negative times and times over 12:00:00 PM on neighboring days. This is not possible in B-COMM. For this reason, times outside the range valid for B-COMM are automatically limited to the valid range upon synchronization.

Please also note that B-COMM supports a maximum of 12 access periods per access time model. Since the configuration of special functions, e.g. public holidays or doors permanently open, is managed differently in B-COMM and HYDRA, more than one B-COMM time slot may result from a single HYDRA access period. This means that the maximum number of 12 time slots may be exceeded even if fewer access periods are configured in HYDRA.



If more than 12 time slots are required in B-COMM, the entire access time model is not transferred to B-COMM, so that this error will be noticed in any case. In addition, this will be recorded in the interface log.

Kaba Benzing does not support any validity period for access time models. For this reason, the currently valid access time model is always transferred to B-COMM. If the access time model changes in the future, this change has to be transferred to the offline components on the relevant day.



As regards changes to the access time models, please take into account that the transfer of access time models to the components via the programmer is time-independent from writing the authorizations on the badge. For this reason, it is almost impossible to change the numbers or meanings of access time models.

3.1.5 Opening hours

B-COMM does not support any [opening hours](#) in the HYDRA sense. The definition of when a badge is granted access depends solely on the access authorizations of the badge.

For B-COMM, the opening hours in HYDRA are used solely to configure Doors permanently open and Office unlocked (see following section).

Kaba Benzing does not support any validity period for opening hours. For this reason, the currently valid opening hours are always processed in the interface to B-COMM. If the access time model of opening hours changes in the future, this change has to be transferred to the offline components on the relevant day.

3.1.6 Opening hours/doors permanently open/office unlocked

It is not possible to combine Doors permanently open and Office unlocked in one access time model in B-COMM. If an access time model with both functions exists in HYDRA, only the Doors permanently open function will be transferred to B-COMM.



The authorization for implementing Office unlocked must be assigned to the badge via the [access authorizations](#) of the access profile in addition to the access time model of HYDRA opening hours.

3.1.7 Multiple access time models for one access group

By assigning several access time models to one badge, it is possible that the badge has several authorizations with different access time models for the same access group. If the validity periods of the authorizations overlap, only the authorization with the lower number of the access time model is written on the badge.



Consequently, the structure of the access time models should ensure that models with a lower number include the longer access periods, and the models with very restricted access periods should be assigned with higher numbers. An access time model granting authorization for 24 hours on every day should be number 1.

3.1.8 Public holidays

For B-COMM, special days and days off are derived from the [public holidays](#) in HYDRA.

HYDRA public holidays of the "Other day off" type become days off in B-COMM. Other types of public holidays become special days.



B-COMM only processes 2 types of public holidays. For this reason, the HYDRA public holiday types *Public holiday* and *Important public holiday* are transferred to B-COMM with the same time authorizations. Consequently, it makes no sense to enter deviating [access periods](#) for the public holiday type *Important public holiday* in the [access time models](#) used in the offline

components.



All future public holidays entered in HYDRA are forwarded to B-COMM and transferred to the offline components via the programmer. After entering the public holidays for a new year, it is necessary to distribute them to all offline components by means of the programmer (provided there are deviating time slots for public holidays in the offline components).

3.2 Data managed in the B-COMM server

3.2.1 Administration area

3.2.1.1 Overview

Only one administration area is managed by the interface from HYDRA. By default, this is the administration area with the number 1. If required, the number can be adjusted by customizing the interface.

HYDRA automatically synchronizes the tabs *Time profiles/TimePro*, *Door(groups)* and *Days off/Special days* of the administration area. The master data of the administration area itself (tabs *Parameters* and *Master*) are not managed by the interface. They must be maintained manually in the B-COMM GUI.

Before the first successful run of the interface, the CardLink administration area has to be created in B-COMM, otherwise synchronization will not be possible.

3.2.1.2 Parameters tab

The settings in the *Parameter* tab are not managed by the interface from HYDRA. They must be edited in B-COMM.

3.2.1.3 Master tab

The settings in the *Master* tab are not managed by the interface from HYDRA. They must be edited in B-COMM.

3.2.1.4 Time profiles/TimePro tab

The data in the *Time profiles/TimePro* tab is maintained completely via the interface from HYDRA and derived from the access time models. Manual modifications in B-COMM are overwritten with the next synchronization from HYDRA.

The *OfficeIndividual* mode is used for Office unlocked in the time profiles of type *TimePro*. As a consequence, it is possible to control Office unlocked for each badge individually.

3.2.1.5 Door (groups) tab

The data in the *Door (groups)* tab is maintained completely via the interface from HYDRA and derived from the access groups and opening hours in HYDRA. Manual modifications in B-COMM are overwritten with the next synchronization from HYDRA.

Since B-COMM does not support any opening hours in the HYDRA sense, the time profile *Default (always)* is assigned to the door groups.

3.2.1.6 Days off/special days tab

The data in the *Days off/special days* tab is maintained completely via the interface from HYDRA and derived from the public holidays in HYDRA. HYDRA public holidays of the "Other day off" type become days off in B-COMM. Other types of public holidays become special days.

Manual modifications in B-COMM are overwritten with the next synchronization from HYDRA.

3.2.2 Components

3.2.2.1 Overview

The components correspond to the accesses in HYDRA. Some settings are synchronized from HYDRA in the components, others do not have any equivalent in HYDRA and must therefore be maintained manually in B-COMM. When synchronizing the components from HYDRA to B-COMM, the following incidents may occur:

1. **The HYDRA access already exists as a component in B-COMM**
The already existing data of the component is superimposed by the data from HYDRA (details are described below).
2. **The HYDRA access HYDRA does not yet exist in B-COMM**
The new component is copied from the first component of B-COMM (sorted by index) and subsequently the data from HYDRA is superimposed (details are described below). If no first component is available, no accesses can be transferred from HYDRA to B-COMM. This is recorded in the log file.
3. **The component in B-COMM no longer exists as an access in HYDRA**
The component is deleted completely from B-COMM.

The key for synchronizing the HYDRA accesses to the components in B-COMM is the door number (*Door (groups)* tab, *Door number* field).

3.2.2.2 Parameters tab

The following fields are synchronized from HYDRA:

Name

The name is composed of the number and the designation of the access. Since B-COMM has restrictions in this field, the following changes are made to the name: Lower case letters are converted into upper case letters, impermissible characters are replaced with minus signs, and the length is limited to 30 characters.

Validation

This field is assigned with the value 3.

Lock opening time (door bolt)

The lock opening time is transferred from the field "Relay time: opener" of the HYDRA access and limited to the valid value range from 2 to 10 seconds.

All other fields must be edited manually in B-COMM.

3.2.2.3 Master tab

The settings in the *Master* tab are not maintained by the interface from HYDRA. They must be edited in B-COMM.

3.2.2.4 Door (groups) tab

Door number

The door number is the key via which HYDRA synchronizes the accesses to the components.

The *Door (groups)* are derived from the HYDRA access groups and accesses.

The tab is transferred completely from HYDRA; manual changes are overwritten with the next synchronization.

3.2.2.5 TimePro tab

The TimePro function is activated by the interface. Time profiles of the TimePro type are derived from the opening hours of the access group of the relevant access.

The tab is completely transferred from HYDRA; manual changes are overwritten with the next synchronization.

3.2.2.6 Days off tab

The days off of a component are derived from the HYDRA public holidays of the *Other day off* type. The access location is considered in this case. A public holiday is valid for the access if the location of the public holiday is empty or coincides with the location of the access group.

3.2.2.7 Special days tab

The special days of a component are derived from the HYDRA public holidays of the *Public holiday and Important public holiday* type. The access location is considered in this case. A public holiday is valid for the access if the location of the public holiday is empty or coincides with the location of the access group.

4 Synchronization of KABA Offline Components

4.1 Overview

The initialization of Kaba offline components (digital cylinders and electronic door mounts) and the synchronization of specific configurations (e.g. access time models and public holidays) are implemented via the Kaba Programmer 1460, which is loaded via the Kaba B-COMM software.

An interface between HYDRA and the B-COMM server automatically synchronizes accesses configured in HYDRA and other access configurations with Kaba B-COMM.

4.2 Configuration of the interface

The technical settings and configurations regarding the interface connection to the B-COMM server are described in a separate document on the [installation](#) of the CardLink function.

4.3 Initiation and process of synchronization

The synchronization of the B-COMM offline components is initiated directly and automatically upon the modification of master data in HYDRA Access Control. Every time, the complete administration area with all sub-elements is synchronized. The following modifications to master data will initiate a synchronization:

- Accesses
- Access groups
- Opening hours
- Access time models
- Access periods
- Public holidays

Since only one synchronization can run at a time, modifications which are made at virtually the same time at various clients in HYDRA will be transferred to the B-COMM server one after the other. This may result in minor synchronization delays.

4.4 Logging

Synchronization runs are recorded in the HYDRA system logs. They are identifiable as BCOMM application. The system logs must be checked regularly for errors.

All started synchronization runs are recorded in the log file `err\zksoffline.pro` of the HYDRA server. This file also indicates when several interface calls are serialized due to the simultaneous modification of master data at different HYDRA clients.

4.5 Access log

The access logs can be read from the offline components by means of the Kaba Programmer and evaluated in B-COMM.

An interface transferring the access logs from B-COMM to HYDRA does not exist.

5 CardLink

5.1 Overview

When using KABA offline components (electronic door mountings and digital cylinders), CardLink allows for loading authorizations onto the badge by using the terminal program AIP or ctwin at PZE terminals. The advantage of this is that in the event of changes to the authorizations, it is not necessary to load the authorizations onto the affected components on site.

When clocking at a PZE terminal, the authorizations on the badge are automatically "validated". A time stamp is updated on the badge, and the authorizations on the badge are subsequently valid for the validation period set in B-COMM. Typically, this period is set to approx. 16 hours, so that there will not be any valid authorizations in the course of the next day for lost badges deactivated in HYDRA, and authorizations also can no longer be validated or loaded at the PZE terminal.

If the PZE terminal detects that the authorizations on the badge are no longer valid, the employee will be informed at the PZE terminal that new authorizations are available and should be loaded onto the badge.

As for the following changes, the system will recognize that the authorizations for offline components have changed:

- **Change in access profile**

A change in the access profile affects all badges to which this access profile is assigned, provided this change affects an access group which includes offline components.

- **Change in access profile assignment**

If the access profile assignments for a badge change and an access configured as an offline component is affected by this.

- **Change in validity period of a badge**

If the validity period of a badge was changed or the badge is deactivated. This may also be initiated by a change of the badge number or the date of leaving in the HR master data.



If several versions with interrupted validity periods exist for a badge in HYDRA, the authorizations are only written onto the badge until the end of the current validity period. Upon expiry of this validity period, the employee is requested to reload the authorizations at the PZE terminal.

5.2 Requirements

The use of CardLink requires memory space on the badge. Legic badges require a separate segment.

A Legic badge must be structured as follows:

- The badge must have two segments: The standard PZE/ZKS access segment and the CardLink segment.
- Each segment must have its own, unambiguous search string.
- The first segment is, by definition, the standard PZE/ZKS access segment with search string + badge data (standard MPDV segment). For this segment, reading access only is required.
- The second segment is the CardLink segment. This segment must have sufficient memory space and allow for both reading and writing access.

At present, CardLink is only available for Legic badges (Legic Prime and Legic Advant).

In addition, a special write-capable reader is required at those PZE terminals where authorizations are to be written on the badge. This is a new LEGIC Advant write-capable reader (with "LGA" in the MPDV product description; example: 382-ILGAL).

5.3 Memory requirements on the legic badge

The required size of the segment is calculated according to the following formula:

$$10 \text{ bytes} + 10 \text{ bytes} \times \text{number of authorized access groups}$$

If a badge is authorized for offline components, which are categorized in 5 different access groups, the memory space requirement is:

$$10 \text{ bytes} + 10 \text{ bytes} \times 5 = 60 \text{ bytes}$$

If the badge is authorized for an access group for several, interrupted periods, another 10 bytes are added for each period.



The required memory space for authorizations has an effect on the duration needed to write authorizations: Approx. 1 second is needed for each 100 bytes. For this reason, you should attempt to summarize the offline component accesses in as few access groups as possible.

5.4 Updating authorizations at the PZE terminal

The PZE terminals load the authorizations for offline components in a cycle of 5 minutes, so that these authorizations can also be validated and/or loaded in the offline case. This cycle duration can be set in seconds in the configuration file `hytnrcfg.ini` for terminals with the terminal program AIP, and in the file `hytnrcfg.bsp` for terminals with the terminal program ctwin, respectively:

```
[ CARDLINK.LST ]  
loadtime=300
```

5.5 Writing authorizations at the PZE terminal

The authorizations of a person are written on the badge at the PZE terminal by using one of the 4 absence reason buttons.

Required program statuses:

Terminal program	Versions
drv_crypt.dll	V# 2.0.0.2
ctwin.exe	V# 7.2.7.19
ctaip.exe .\\packets\\pzezks72.dll	V# 2.0.3.7 V# 2.0.1.22

5.5.1 Terminal configuration at the console

The CardLink function is configured in the [terminal configuration](#) in the tab "HR functions". The writing of authorizations on the badge can be configured with the "Absence reason" **CL** and an appropriate text on an absence reason button.

5.5.2 Terminal configuration

For writing on badges, a reader-specific DLL has to be configured at the terminal. At present, the following DLL are available:

DLL	Description
drv_crypt.dll	Driver for LEGIC Advant reader (e.g. 382-ILGAL, CTB-LGALU, CT-LGALTU)

The configuration is performed at the terminal in the file **ctaip.ini** for AIP terminals, and in the file **ctwin.ini** for terminals with the ctwin terminal program, respectively.

Driver activation (x stands for the number of the serial interface):

```
[COMPORTS]
COMx=drv_crypt
```

Activating the CardLink functionality:

```
[COMPORTS-PARAM]
DRV_CRYPT-PARAM=SETTINGS=CL|
```

Customer-specific configuration (example):

```
[COMPORTS-PARAM]
DRV_CRYPT-PARAM=SETTINGS=CL|SEARCHSTRING=2C2D2E000000|STARTADDRESS=6|
```


Depending on the structure of the badges used, a customer-specific configuration may also be required. Primarily, this includes settings for the CardLink segment: search string, segment size and data start address.



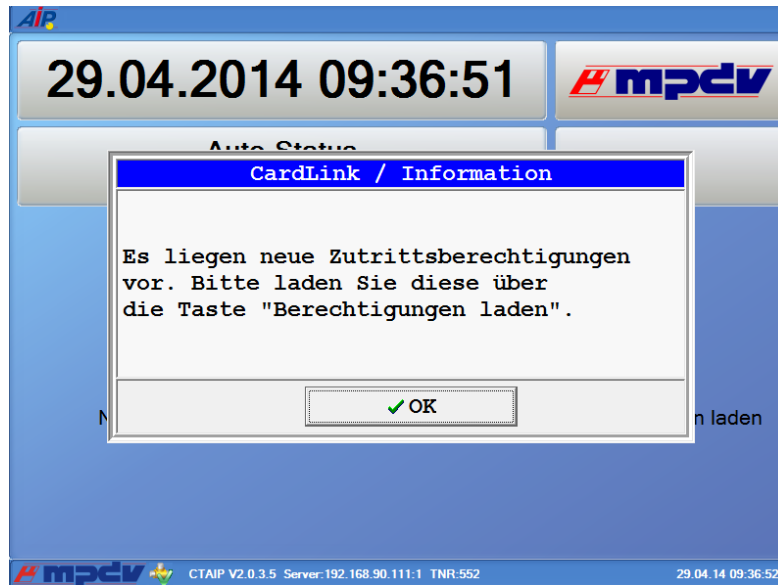
These settings have an impact on the reader behavior. These entries should be made carefully, since they have an effect on the overall behavior of the CardLink application. Adaptations should only be made after consultation with MPDV.

The parameters described below can be set within the parameter string "DRV_CRYPT-PARAM" in order to implement the customer-specific configuration. If several parameters are used, they are to be separated by "|" (Pipe).

Parameter	Description
SEARCHSTRING	Search string of the CardLink segment The default is "2C2D2E000000"
STARTADDRESS	This indicates the start address on the badge within the access segment. Usually immediately behind the search string. Example: The entry 3 corresponds to the fourth byte as start address (counting starts at 0). The authorizations are consequently written as from the fourth byte. The default value is 6.

5.5.3 Process at the terminal

New rights are available for a badge:



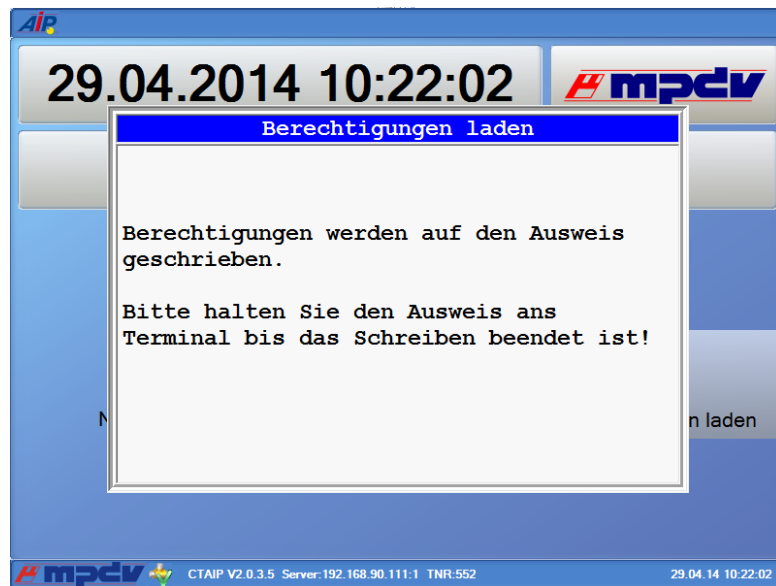
The function "Load authorizations" is selected via the relevant function key.



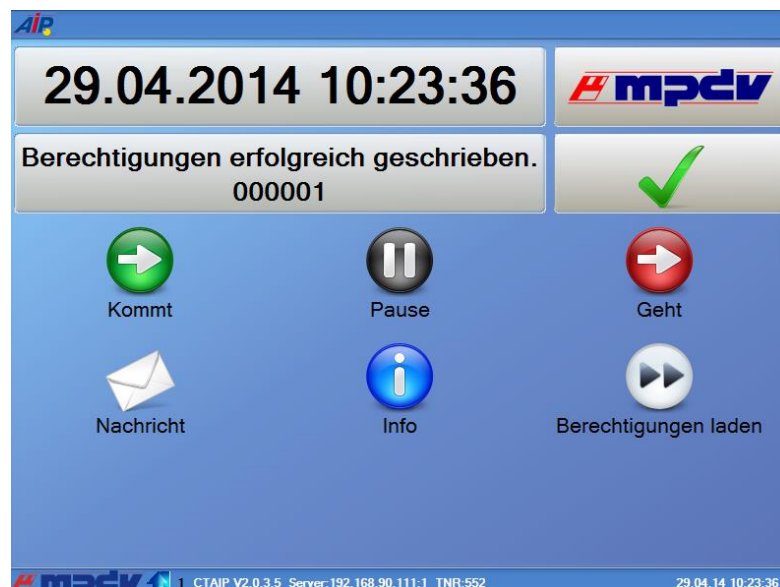
Please note:

The screenshots show the display of the AIP terminal when loading authorizations on a badge. The display at CTWIN is comparable in terms of the contents.

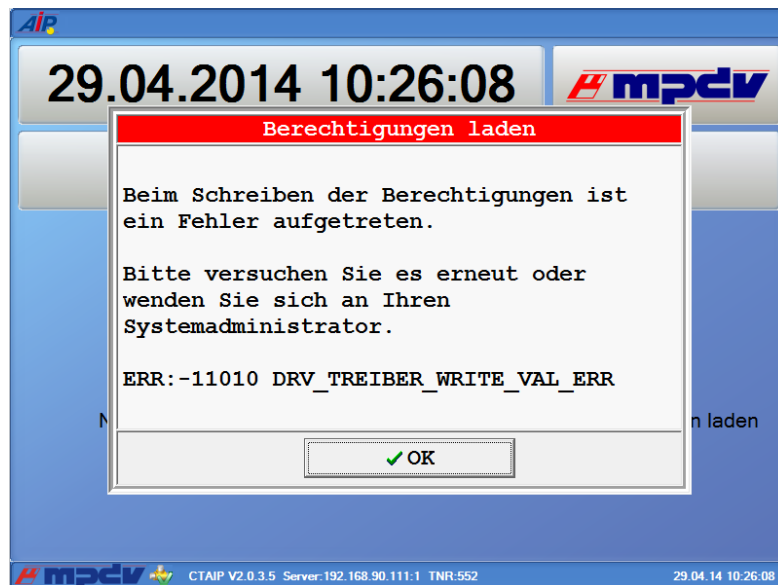
The following note is displayed when authorizations are written on a badge:



After the successful writing of authorizations, the note is closed and successful processing is indicated as follows, as when clocking:



In the case of an error, a message window opens, which can be closed by selecting "OK". The contents depend on the error occurred.



Possible errors are:

Error	Description
-11008	DRV_TREIBER_ERROR_WRITEDATA Error while writing the badge data. Possible causes: - Badge removed from reader during writing. - Badge does not have sufficient memory space. - Communication with reader and/or badge disrupted.
-11009	DRV_TREIBER_TIMEOUT_WRITEDATA Timeout while writing the badge data. Possible causes: - Communication with reader and/or badge disrupted.
-11010	DRV_TREIBER_WRITE_VAL_ERR Error during validation of badge. Possible causes: - Badge removed from reader during writing. - Communication with reader and/or badge disrupted.
-11011	DRV_TREIBER_WRITE_DEVAL_ERR Error during invalidation of badge. Possible causes: - Badge removed from reader during writing. - Communication with reader and/or badge disrupted.
-11013	DRV_TREIBER_WRITE_DATA_SEG_TO_SMALL_VALID The data to be written does not fit into the CardLink segment. Possible causes:

	- The CardLink segment is too small. - Authorizations have to be optimized.
-11014	DRV_TREIBER_WRITE_DATA_SEG_TO_SMALL_NOT_VALID The data to be written does not fit into the CardLink segment. Possible causes: - The CardLink segment is too small. Authorizations have to be optimized.

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